SECV2113 HUMAN-COMPUTER INTERACTION

SECTION 01

SEMESTER 2 2025/2026

GROUP 3 - FALSE 9

GROUP ASSIGNMENT

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INTRODUCTION

This report documents the usability inspection of a newly developed bus tracking system prototype. The evaluation aims to identify usability issues, assess their impact on the user experience, and provide actionable recommendations for improvement.

The inspection was conducted using a Heuristic Evaluation (HE) methodology, where the evaluator assumes the role of a usability expert. The prototype's user interface was systematically reviewed against Nielsen's 10 usability heuristics.

- Any identified usability violations were recorded and assigned a severity rating ranging from 1 to 4.
- The ratings scale classifies issues from a "Cosmetic issue" (Severity 1) that should be fixed if time permits, up to a "Catastrophic issue" (Severity 4) that prohibits users from completing their tasks and requires immediate modification.
- Following the heuristic evaluation, the findings were incorporated into a User Journey Map to visualize the user's emotional state, highlight specific pain points during interaction, and propose targeted design recommendations.

1.2 Prototype Description

The evaluated prototype is a comprehensive campus bus tracking system designed to simplify and optimize student commutes. Its core features include:

- **Real-Time Tracking and Route Integration:** The system provides live bus positions, estimated arrival times, traffic conditions, and a multi-stop route planner. It integrates both public and campus buses into a single platform, supporting Sustainable Development Goals (SDGs) 9, 11, and 13 by promoting accessible, smart, and eco-friendly digital infrastructure.
- **Personalised Scheduling and Notifications:** Users can schedule trips in advance using an integrated calendar. The interface features an emotionally adaptive virtual assistant that provides spoken or pop-up notifications and adjusts its responses based on user sentiment (e.g., anxiety or frustration).
- **User Feedback Dashboard:** A centralized hub where users can report issues (like delays or overcrowding) via a submission form that generates a tracking ID, while also viewing broader service updates and system announcements.

1.3 Persona, Scenario, and User Tasks

To ensure the evaluation remains grounded in real-world use, the inspection was guided by a specific target user profile, a practical scenario, and defined user tasks.

Persona: Aiman

Aiman is a 21-year-old tech-savvy student at UTM. Rushing between classes and social activities, he relies heavily on real-time tracking apps to make fast decisions about his campus movements. He expects instant updates, hates cluttered interfaces, and gets highly frustrated when app inaccuracies disrupt his tight schedule.

Scenario

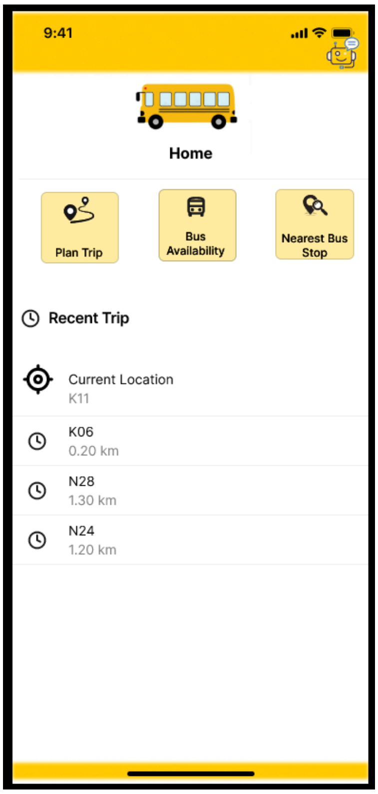
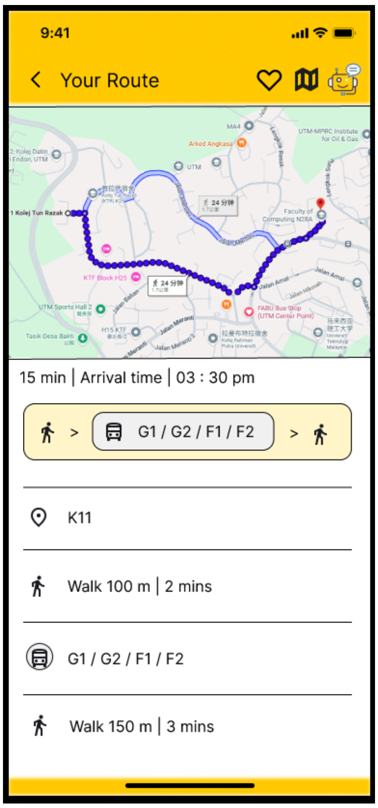
Aiman is waiting at the UTM bus stop early in the morning, needing to reach his lecture room before class begins. He opens the Fourmula 1 app to check the bus arrival time. Navigating to "Track Bus," he sees that his bus is arriving in exactly 3 minutes. Relieved, he decides to wait rather than walk. The bus arrives on schedule, allowing Aiman to board easily, reach his class on time, and appreciate the app's efficiency.

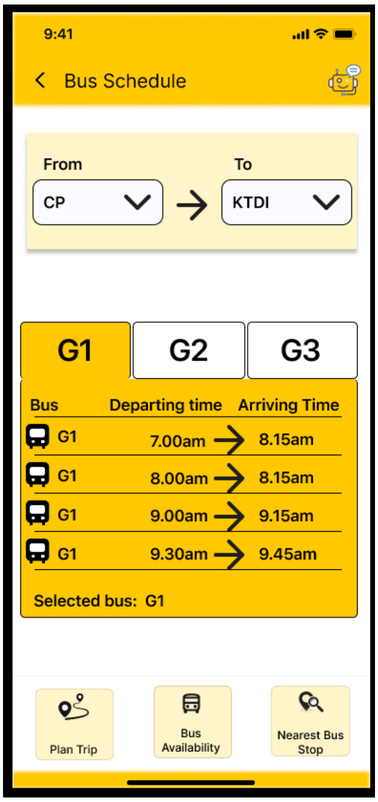
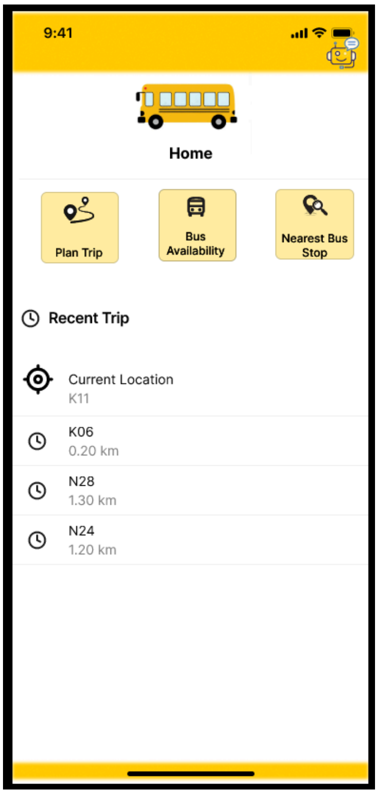
User Tasks

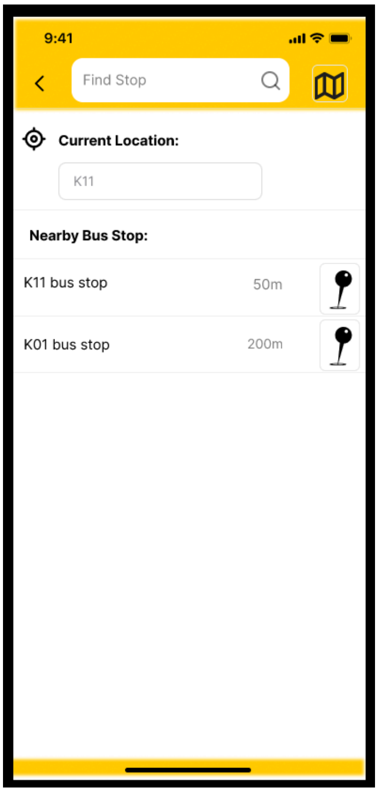
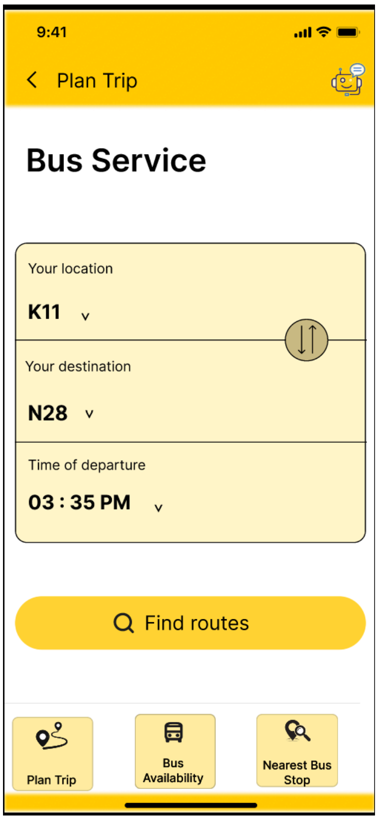
Based on Aiman's goals and scenario, the evaluation focused on the following three primary user tasks:

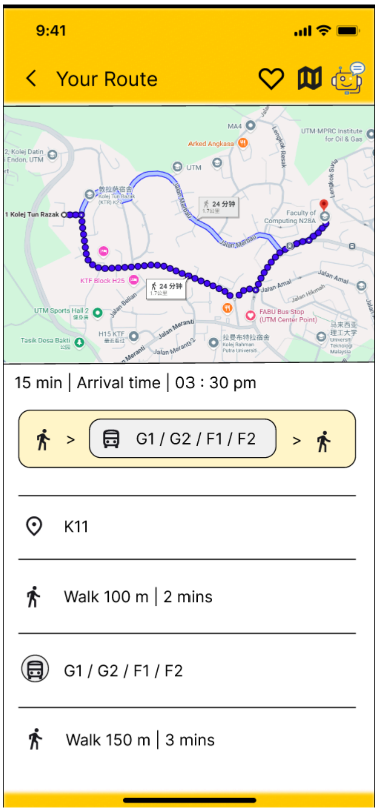
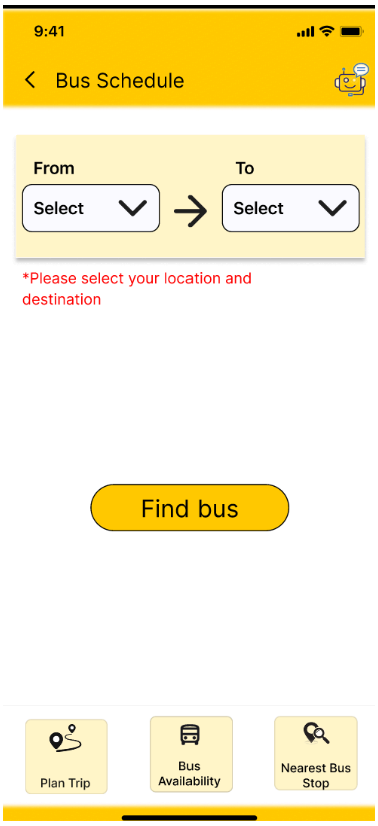
1. **Task 1 :** Track Bus Availability
2. **Task 2 :** Find Nearest Station.
3. **Task 3 :** Plan Trip and Save Trip

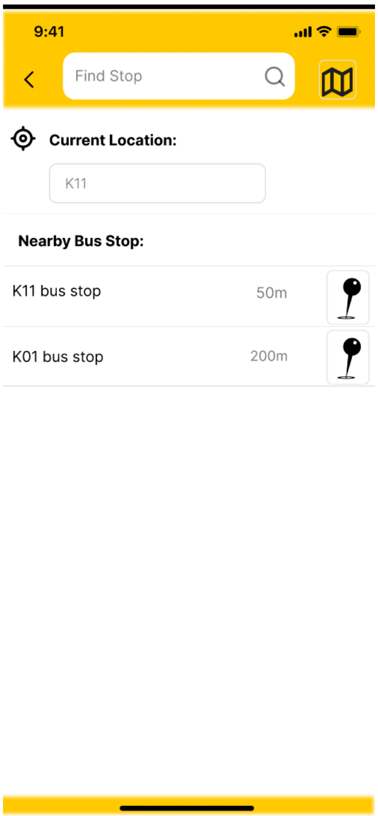
HE RESULT

Prototype image	Identified Issue	Heuristic and Severity
	The "Current Location" item (paired with a target icon) is placed directly underneath the "Recent Trip" heading and clock icon, even though a user's current location is not a past trip.	H8: Aesthetic and minimalist design S2: Minor issue
	There is a discrepancy in the estimated travel time. The map graphic displays a tag indicating "24 min", while the text summary immediately below it states "15 min".	H4: Consistency and standards S3: Major issue

 	The page header is clearly titled "Bus Schedule", but the navigation button used to represent this feature in the menu is labeled "Bus Availability".	H4: Consistency and standards S2: Minor issue

	The "Current Location" field displaying "K11" is styled with a bounding box, making it look exactly like an editable text input field, confusing the user on whether they can manually type a location or if it is a static label.	H4: Consistency and standards S2: Minor issue
	In the bottom navigation bar, all three menu items (Plan Trip, Bus Availability, and Nearest Bus Stop) use the exact same background color, border, and visual style. There is no clear visual highlight or active state indicator to show the user that they are currently on the "Plan Trip" page.	H1: Visibility of system status S2: Minor issue

	Under the arrival time summary, the app bundles four completely different bus lines together as a single step: "G1 / G2 / F1 / F2". It fails to specify which exact bus is coming next, which one the user should wait for, or if they all follow identical paths to the same stop.	H2: Match between system and real world S2: Minor issue
	The red error text, "Please select your location and destination", is hardcoded directly onto the screen before the user has even had an opportunity to click anything or attempt to submit the form.	H5: Error prevention S2: Minor issue

	The map pin on the right side of the list K11 bus stop and K01 bus stop, this are poorly cropped or styled. The circular head of the pin icon is cut off at the top border of its container box, and the visual styling looks outdated and unaligned. Furthermore, there is a clear lack of signifiers, it is completely unclear if clicking the pin icon does something different (like opening a map preview) versus clicking the row text itself.	H8: Aesthetic and minimalist S1: Cosmetic issue

USER JOURNEY MAP

 **Aiman**	**Scenario** a student who uses the UTM Fleet app to check real-time arrival updates	**Goals** • to get instant, accurate bus tracking updates. • Make a quick decision on whether to wait or walk. • Arrive at the lecture room on time without feeling stressed	
Phase	Track Bus Availability	Find Nearest Station.	Plan Trip and Save Trip
Actions	1. pulls out his phone 2. opens the tracking app 3. selects the bus schedule feature 4. manually enters his "From" and "To" information **Touchpoints:** App Home Screen, Navigation Menu, Location Input Fields. **Channels:** Mobile App (Fourmula 1 Bus).	1. pulls out his phone 2. opens the tracking app 3. selects the nearest bus stop feature 4. the current location will be manually entered by the user. **Touchpoints:** App Home Screen,Nearest Bus stop icon , Location Input Fields. **Channels:** Mobile App (Fourmula 1 Bus).	1. pull out his phone 2. selects plan trip feature 3. Manually enter his location, destination, and time of departure. 4. selects the find route button 5. selects the heart icon 6. manually enters his preferred save name **Touchpoints:** App Home Screen, Navigation Menu, Location, Destination and TIme Input Fields, Route Details Screen
Emotion	 “It is so easy to find the bus schedule.”	 “That would be very simple for me”	 “ The save feature is interesting “

Recommendations	Implement a GPS auto-detect feature that automatically fills in the user's current location at the bus stop	Change the thumbtack icon to a standard teardrop map pin icon or a small bus icon to align with the functionality of public transportation apps	Replace the heart icon in the top header with a text-based “Save Trip” button directly below the arrival time summary.

References

https://sites.google.com/view/bpostgroup5/FourmulaTeam

https://www.figma.com/proto/ZkKjAaTobz2bTz0sN6hTMk/HCI-Wireframe--Fourmula-?node-id=0-1&t=lzvbHz0OzjLVghJe-1